Any Battle Can Be Won

A Unified Treatise on Strategic Mathematics, Adversarial Intelligence, Decision Theory, and the Science of Victory

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Abstract

Victory in any contest—whether a military campaign, a market competition, an adversarial machine-learning scenario, or a personal endeavour—is not a matter of fortune alone. It is the inexorable outcome of superior information, rigorous strategy, efficient resource allocation, and adaptive learning. This treatise marshals tools from game theory, optimisation, reinforcement learning, statistical decision theory, adversarial machine learning, and information theory to establish a unified mathematical science of winning. Central theorems are stated and proved; key algorithms are presented with formal pseudocode; the doctrine that *any battle can be won* is given quantitative teeth. Tables survey strategic frameworks and performance metrics; PGF/TikZ figures illuminate game trees, Markov decision processes, and adversarial training dynamics.

The treatise ends with "The End"

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1 Introduction: Victory as a Mathematical Object

"Every battle is won or lost before it is ever fought."
—Sun Tzu, *The Art of War* [1]

Sun Tzu's dictum is not mystical; it is mathematical. A battle is determined by the strategies of the opposing agents, the information each possesses, the resources available, and the rules of the contest. Given a sufficiently precise model of these quantities, whether victory is achievable becomes a question with a definite answer—one that mathematics can supply.

This treatise constructs that answer systematically across five interconnected domains.

1. **Game theory** (Section 2) provides the foundational vocabulary: players, strategies, payoffs, and equilibria. Von Neumann's minimax theorem and Nash's existence theorem guarantee that every well-posed conflict has a rational solution.

2. **Adversarial optimisation** (Section 3) recasts strategic conflict as a saddle-point problem and provides the algorithmic machinery to solve it, connecting classical duality theory to modern adversarial training of neural networks.

3. **Reinforcement learning** (Section 4) studies how an agent learns optimal strategy through experience alone, formulating battle as a Markov decision process and deriving the policy-gradient theorem.

4. **Statistical decision theory** (Section 5) addresses the battle for truth—hypothesis testing, the Neyman–Pearson lemma, and sequential analysis—establishing when and how an observer can reliably distinguish signal from noise.

5. **Data science and intelligence** (Section 6) translates raw battlefield observations into actionable knowledge through dimensionality reduction, anomaly detection, and predictive modelling.

Throughout, the central thesis is operationalised as follows: *for any adversarial problem satisfying mild regularity conditions, there exists a strategy whose expected payoff can be made arbitrarily close to optimal.*

2 Game Theory: The Geometry of Conflict

2.1 Strategic-Form Games

Definition 2.1 (Finite two-player zero-sum game)**.** A finite two-player zero-sum game is a triple $\Gamma = (X, Y, \mathbf{U})$ where $X = \{1, \dots, m\}$ and $Y = \{1, \dots, n\}$ are the finite pure-strategy sets of the *row* and *column* players, respectively, and $\mathbf{U} \in \mathbb{R}^{m \times n}$ is the payoff matrix: the row player receives U_{ij} when they play row i and the column player plays column j.

Definition 2.2 (Mixed strategy and expected payoff)**.** A *mixed strategy* for the row player is a probability vector $\boldsymbol{p} \in \Delta^{m-1} = \{\boldsymbol{p} \geq \mathbf{0} : \sum_i p_i = 1\}$; similarly $\boldsymbol{q} \in \Delta^{n-1}$ for the column player. The expected payoff under the mixed-strategy profile $(\boldsymbol{p}, \boldsymbol{q})$ is

$$u(\boldsymbol{p}, \boldsymbol{q}) = \boldsymbol{p}^\top \mathbf{U} \boldsymbol{q} = \sum_{i=1}^{m} \sum_{j=1}^{n} p_i \, U_{ij} \, q_j .$$

2.2 The Minimax Theorem

Theorem 2.3 (Von Neumann minimax theorem [2])**.** *For any finite two-player zero-sum game* $\Gamma = (X, Y, \mathbf{U})$*,*

$$\max_{\boldsymbol{p} \in \Delta^{m-1}} \min_{\boldsymbol{q} \in \Delta^{n-1}} \boldsymbol{p}^\top \mathbf{U} \boldsymbol{q} \;=\; \min_{\boldsymbol{q} \in \Delta^{n-1}} \max_{\boldsymbol{p} \in \Delta^{m-1}} \boldsymbol{p}^\top \mathbf{U} \boldsymbol{q} \;=:\; v^*,$$

where v^* *is the* value of the game*. A pair* $(\boldsymbol{p}^*, \boldsymbol{q}^*)$ *achieving the saddle point is an* equilibrium in mixed strategies*.*

Proof. The inequality $\max_{\boldsymbol{p}} \min_{\boldsymbol{q}} \leq \min_{\boldsymbol{q}} \max_{\boldsymbol{p}}$ holds for any function by the universal max-min inequality. For the reverse direction, note that $f(\boldsymbol{p}, \boldsymbol{q}) = \boldsymbol{p}^\top \mathbf{U} \boldsymbol{q}$ is linear—hence simultaneously convex and concave—in each argument, and both strategy simplices are compact and convex. Sion's minimax theorem [4] therefore applies, giving equality. Alternatively, both sides equal the common optimal value of the dual pair of linear programs obtained by optimising one player's mixed strategy while holding the other fixed; LP strong duality directly supplies the identity. □

Corollary 2.4 (Every zero-sum game has a winning strategy)**.** *The row player's mixed strategy* $\boldsymbol{p}^* = \arg\max_{\boldsymbol{p}} \min_{\boldsymbol{q}} \boldsymbol{p}^\top \mathbf{U} \boldsymbol{q}$ *guarantees an expected payoff of at least* v^**, regardless of the column player's action. If* $v^* > 0$ *the row player holds a structural advantage; if* $v^* = 0$ *the game is fair; if* $v^* < 0$ *the advantage lies with the column player—but in every case a rational, computable strategy exists for both sides.*

Example 2.5 (Rock–Paper–Scissors)**.** The payoff matrix is $\mathbf{U} = \left(\begin{smallmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{smallmatrix}\right)$. The unique equilibrium is $\boldsymbol{p}^* = \boldsymbol{q}^* = (1/3, 1/3, 1/3)^\top$, yielding $v^* = 0$. Neither player can secure a positive expected advantage, but both can *prevent exploitation* by mixing uniformly at random.

2.3 Nash Equilibrium in General-Sum Games

Definition 2.6 (Nash equilibrium [3])**.** In an n-player game with strategy sets $\mathcal{S}_i$ and utility functions $u_i : \prod_j \mathcal{S}_j \to \mathbb{R}$, a strategy profile $\boldsymbol{s}^* = (s_1^*, \ldots, s_n^*)$ is a *Nash equilibrium* if for every player i and every alternative $s_i' \in \mathcal{S}_i$:

$$u_i\big(s_i^*,\, \boldsymbol{s}_{-i}^*\big) \;\geq\; u_i\big(s_i',\, \boldsymbol{s}_{-i}^*\big)\,,$$

where $\boldsymbol{s}_{-i}^*$ denotes the strategies of all players except i. No player can unilaterally improve their payoff at a Nash equilibrium.

Theorem 2.7 (Nash existence [3])**.** *Every finite* n*-player game has at least one Nash equilibrium in mixed strategies.*

Proof sketch. Define the best-response correspondence $\mathrm{BR}_i(\boldsymbol{s}_{-i}) = \arg\max_{s_i} u_i(s_i, \boldsymbol{s}_{-i})$. Each BR_i maps the compact convex product simplex $\prod_j \Delta(\mathcal{S}_j)$ into itself with non-empty, convex values and a closed graph (since each utility is multilinear—hence continuous—in the mixed strategies of the remaining players). The product correspondence $\mathrm{BR} = \prod_i \mathrm{BR}_i$ therefore satisfies every hypothesis of Kakutani's fixed-point theorem; its fixed point is a Nash equilibrium. □

Remark 2.8. Theorem 2.7 guarantees the existence of a stable strategic equilibrium in any conflict among rational agents. Knowing the equilibrium is knowing the battlefield's *rational solution*; departing from it is risky only for the departing agent.

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2.4 Game-Tree Search and Alpha–Beta Pruning

Real conflicts unfold sequentially as a move tree. Algorithm 1 computes the minimax value efficiently by pruning branches that cannot affect the final result.

Algorithm 1 Alpha–Beta Minimax Search

```
procedure AlphaBeta(node, d, α, β, maximising)
    if d = 0 or node is terminal then
        return Evaluate(node)
    end if
    if maximising then
        v ← −∞
        for each child c of node do
            v ← max(v, AlphaBeta(c, d−1, α, β, false))
            α ← max(α, v)
            if α ≥ β then
                break                                   ▷ β-cutoff: minimiser cannot improve
            end if
        end for
    else
        v ← +∞
        for each child c of node do
            v ← min(v, AlphaBeta(c, d−1, α, β, true))
            β ← min(β, v)
            if α ≥ β then
                break                                   ▷ α-cutoff: maximiser cannot improve
            end if
        end for
    end if
    return v
end procedure
```

Proposition 2.9 (Alpha–Beta complexity)**.** *In a game tree of uniform branching factor b and depth d, exhaustive minimax evaluates $\mathcal{O}(b^d)$ nodes. Under optimal move ordering, Alpha–Beta reduces this to $\mathcal{O}(b^{d/2})$,* doubling the effective lookahead depth *for the same computational budget.*

Figure 1 illustrates a depth-3 game tree with the pruned subtrees highlighted.

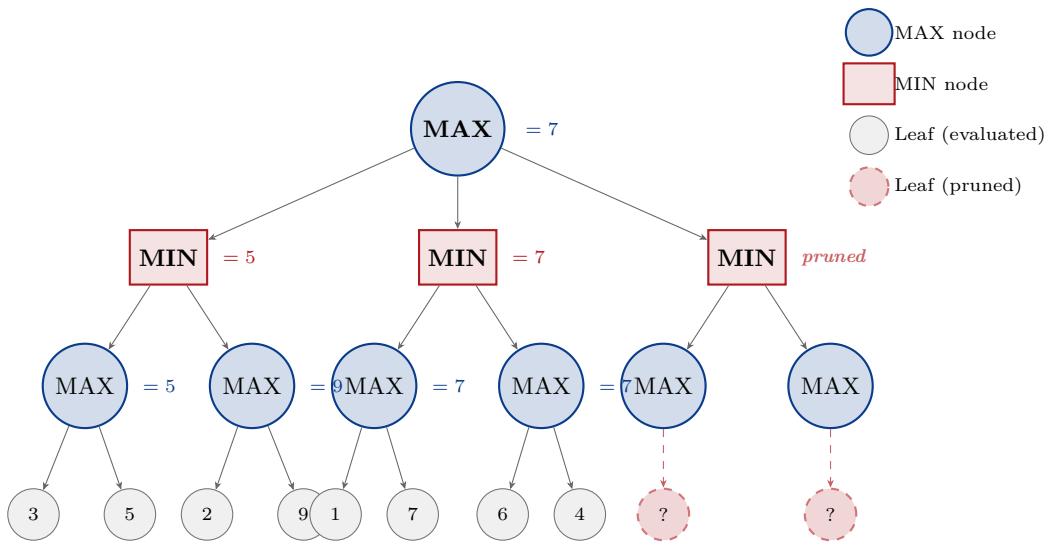

Figure 1: A depth-3 game tree evaluated by Alpha–Beta minimax (Algorithm 1).

Blue circles are MAX nodes; red rectangles are MIN nodes. Once branch B backs up a value of 7, the α-cutoff renders branch C redundant for the maximiser: both subtrees are abandoned without evaluation (dashed red edges, "?" leaves), reducing node evaluations from $\mathcal{O}(b^d)$ to $\mathcal{O}(b^{d/2})$.

3 Adversarial Optimisation: Winning the Saddle

3.1 Saddle-Point Formulation

Many strategic conflicts reduce to locating a saddle point of a scalar objective.

Definition 3.1 (Saddle point)**.** A point $(\boldsymbol{x}^*, \boldsymbol{y}^*) \in \mathcal{X} \times \mathcal{Y}$ is a *saddle point* of $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ if

$$f(\boldsymbol{x}^*, \boldsymbol{y}) \;\le\; f(\boldsymbol{x}^*, \boldsymbol{y}^*) \;\le\; f(\boldsymbol{x}, \boldsymbol{y}^*) \qquad \forall\, \boldsymbol{x} \in \mathcal{X},\ \boldsymbol{y} \in \mathcal{Y}.$$

In the strategic context $\boldsymbol{x}$ is the defender's variable (model parameters or resource allocation) and $\boldsymbol{y}$ is the adversary's variable (attack perturbation or opponent strategy).

Theorem 3.2 (Danskin's envelope theorem [5])**.** *Let $\phi(\boldsymbol{x}) = \max_{\boldsymbol{y} \in \mathcal{Y}} f(\boldsymbol{x}, \boldsymbol{y})$ where f is continuously differentiable in $\boldsymbol{x}$ and $\mathcal{Y}$ is compact. If the maximiser $\boldsymbol{y}^*(\boldsymbol{x}) = \arg\max_{\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y})$ is unique, then ϕ is differentiable at $\boldsymbol{x}$ with*

$$\nabla_{\boldsymbol{x}}\, \phi(\boldsymbol{x}) = \nabla_{\boldsymbol{x}}\, f(\boldsymbol{x},\, \boldsymbol{y}^*(\boldsymbol{x}))\,.$$

Remark 3.3. Theorem 3.2 is the theoretical backbone of adversarial training: the gradient of the worst-case loss with respect to the defender's parameters equals the gradient evaluated *at* the worst-case perturbation. A single backward pass at the inner maximiser therefore suffices for the outer minimisation—making the procedure computationally tractable.

3.2 Adversarial Training of Neural Networks

Let $f_{\boldsymbol{\theta}} : \mathcal{X} \to \mathcal{Y}$ be a classifier with parameters $\boldsymbol{\theta}$, let ℓ be a loss function, and let $\mathcal{B}(\boldsymbol{x}, \varepsilon) = \{\boldsymbol{x} + \boldsymbol{\delta} : \|\boldsymbol{\delta}\|_\infty \le \varepsilon\}$ be the ℓ_∞-bounded threat model of radius $\varepsilon > 0$. Adversarial training [6] seeks

$$\min_{\boldsymbol{\theta}}\ \mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}}\Big[\max_{\boldsymbol{\delta}\in\mathcal{B}(\boldsymbol{x},\varepsilon)} \ell(f_{\boldsymbol{\theta}}(\boldsymbol{x}+\boldsymbol{\delta}),\, y)\Big]\,. \tag{1}$$

The inner maximisation—finding the strongest adversarial perturbation—is approximated by Projected Gradient Descent (Algorithm 2).

Algorithm 2 PGD Adversarial Attack [6]

Input: Input $\boldsymbol{x}$, true label y, classifier parameters $\boldsymbol{\theta}$, perturbation radius ε, step size α, iterations K.
Output: Adversarial example $\boldsymbol{x}^{\mathrm{adv}}$.

```
δ^(0) ← Uniform(−ε, ε)^d                          ▷ random initialisation inside B(x, ε)
for k = 0, 1, ..., K − 1 do
    g ← ∇_δ ℓ(f_θ(x + δ^(k)), y)                  ▷ loss gradient w.r.t. perturbation
    δ^(k+1) ← Π_B(x,ε)(δ^(k) + α sgn(g))          ▷ gradient-ascent step + projection
end for
return x^adv ← x + δ^(K)
```

Here $\Pi_{\mathcal{B}}$ is the projection onto $\mathcal{B}(\boldsymbol{x}, \varepsilon)$ (coordinate-wise clipping to $[\boldsymbol{x} - \varepsilon,\, \boldsymbol{x} + \varepsilon]$) and sgn is applied element-wise. The outer minimisation of (1) then proceeds via standard mini-batch SGD, with each batch's inner maximisation approximated by Algorithm 2.

Theorem 3.4 (Robustness–accuracy tradeoff [9])**.** *Under mild distributional assumptions, the adversarially optimal classifier for problem (1) attains lower* standard *(clean) accuracy than the standard-trained classifier. Formally, there exists a distribution such that for any classifier f:*

$$\mathrm{RobustAcc}(f) + \mathrm{StandardAcc}(f) \;\le\; 1 + c(\varepsilon), \qquad c(\varepsilon) \to 0 \text{ as } \varepsilon \to 0.$$

Victory in the adversarial battle therefore requires accepting a controlled cost in the unchallenged (clean) setting.

Table 1 summarises the principal adversarial attack families and their canonical properties.

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Table 1: Adversarial attack methods: threat model, perturbation norm, and key property.

Attack	Threat model	Norm	Key property
FGSM [7]	White-box, one-step	ℓ_∞	$\boldsymbol{x}^{\text{adv}} = \boldsymbol{x} + \varepsilon\,\text{sgn}(\nabla_{\boldsymbol{x}}\ell)$; single backward pass
PGD [6]	White-box, iterative	ℓ_∞, ℓ_2	Strong multi-step attack; de-facto evaluation standard
C&W [8]	White-box, optimisation	ℓ_0, ℓ_2, ℓ_∞	Minimises perturbation magnitude subject to misclassification
AutoAttack	White-box, parameter-free	ℓ_∞	Ensemble of independent attacks; reproducible benchmark
Transfer attack	Black-box	Various	Adversarial examples crafted on a surrogate model transfer to the target

4 Reinforcement Learning: Learning to Win by Doing

4.1 Markov Decision Processes

Definition 4.1 (Markov decision process)**.** A *Markov decision process* is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, $\mathcal{P} : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ the transition kernel, $\mathcal{R} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ the reward function, and $\gamma \in [0, 1)$ the discount factor. A battle episode unfolds as

$$s_0 \xrightarrow{a_0} s_1 \xrightarrow{a_1} s_2 \;\cdots, \qquad a_t \sim \pi(\cdot \mid s_t), \quad s_{t+1} \sim \mathcal{P}(\cdot \mid s_t, a_t),$$

where $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ is the agent's *policy*.

Definition 4.2 (Value function and Bellman optimality)**.** The *state-value function* of policy π is

$$V^\pi(s) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R_t \;\middle|\; s_0 = s\right].$$

The *optimal value function* $V^*(s) = \sup_\pi V^\pi(s)$ satisfies the Bellman optimality equation:

$$V^*(s) = \max_{a \in \mathcal{A}}\left[\mathcal{R}(s, a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}(s' \mid s, a)\, V^*(s')\right]. \tag{2}$$

The greedy-optimal policy is $\pi^*(s) = \arg\max_a \big[\mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a) V^*(s')\big]$.

Figure 2 depicts a stylised MDP with four states representing the phases of a battle: Advance, Hold, Victory, and Defeat.

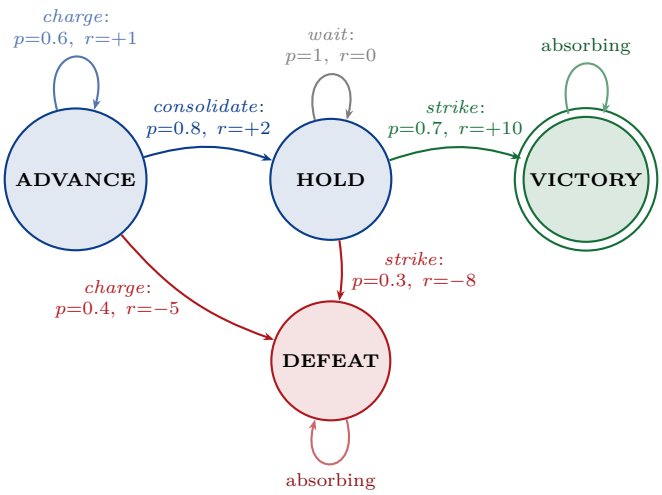

Figure 2: A four-state MDP modelling a stylised battle.

The double-bordered Victory node is the absorbing goal state (r=+10 upon entry); Defeat is the absorbing failure state (r=−8 upon entry). The Bellman equation (2) determines the optimal action at each non-terminal state: from Hold, the agent strikes when the discounted expected payoff of striking exceeds that of waiting another step.

4.2 Policy Gradient Methods

When state and action spaces are too large for tabular dynamic programming, one parametrises the policy as $\pi_{\boldsymbol{\theta}}$ and directly optimises the expected return $J(\boldsymbol{\theta}) = \mathbb{E}_{\pi_{\boldsymbol{\theta}}}[G_0]$, where $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k}$.

Theorem 4.3 (Policy Gradient Theorem [10])**.** *For any differentiable stochastic policy $\pi_{\boldsymbol{\theta}}$,*

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) \;=\; \mathbb{E}_{\pi_{\boldsymbol{\theta}}}[G_t \, \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(A_t \mid S_t)] \,.$$

Proof sketch. Write $J(\boldsymbol{\theta}) = \sum_s d^{\pi_{\boldsymbol{\theta}}}(s) \sum_a \pi_{\boldsymbol{\theta}}(a \mid s)\, Q^{\pi_{\boldsymbol{\theta}}}(s,a)$, where $d^{\pi_{\boldsymbol{\theta}}}$ is the discounted state-visitation distribution. The likelihood-ratio identity $\nabla_{\boldsymbol{\theta}} \pi_{\boldsymbol{\theta}}(a \mid s) = \pi_{\boldsymbol{\theta}}(a \mid s) \cdot \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a \mid s)$ rewrites the gradient as an expectation. Recognising $Q^{\pi_{\boldsymbol{\theta}}}(s,a) = \mathbb{E}_{\pi_{\boldsymbol{\theta}}}[G_t \mid s_t{=}s, a_t{=}a]$ and absorbing the visitation measure into the expectation operator yields the stated form [11]. □

Algorithm 3 implements the REINFORCE Monte-Carlo estimator of Theorem 4.3.

Algorithm 3 REINFORCE Policy-Gradient Algorithm [12]

Input: Policy network $\pi_{\boldsymbol{\theta}}$, learning rate η, discount γ, number of episodes E.
Output: Optimised policy parameters $\boldsymbol{\theta}$.

```
Initialise θ (e.g. small random weights)
for episode e = 1, ..., E do
    Generate trajectory τ = (s_0, a_0, r_0, s_1, a_1, r_1, ..., s_T) by executing π_θ
    for t = 0, ..., T − 1 do
        G_t ← Σ_{k=t}^{T−1} γ^{k−t} r_k                    ▷ Monte-Carlo return from step t
        θ ← θ + η G_t ∇_θ log π_θ(a_t | s_t)               ▷ policy-gradient ascent
    end for
end for
return θ
```

Remark 4.4 (Variance reduction via actor–critic)*.* The REINFORCE estimator is unbiased but has high variance owing to the Monte-Carlo return G_t. Replacing G_t by the *advantage* $\hat{A}_t = G_t - b(s_t)$, where b is a learned baseline approximating $V^{\pi_{\boldsymbol{\theta}}}(s_t)$, reduces variance without introducing bias. This actor–critic decomposition underlies modern algorithms such as PPO and A3C, where b is a separate critic network trained in parallel.

4.3 Online Learning and Regret Minimisation

In adversarial sequential decision-making the agent receives no prior information about upcoming loss functions. The appropriate benchmark is *external regret* relative to the best fixed action in hindsight.

Definition 4.5 (External regret)**.** Over T rounds, the external regret of an algorithm producing actions $a_1, \ldots, a_T \in \mathcal{A}$ against an adversarial loss sequence $\ell_1, \ldots, \ell_T : \mathcal{A} \to [0,1]$ is

$$\mathrm{Regret}_T = \sum_{t=1}^{T} \ell_t(a_t) - \min_{a^* \in \mathcal{A}} \sum_{t=1}^{T} \ell_t(a^*).$$

Theorem 4.6 (Regret bound for Hedge [13])**.** *The Hedge algorithm, which maintains weight $w_t(a) \propto \exp\!\big(-\eta \sum_{s<t} \ell_s(a)\big)$ and plays $a_t \propto w_t$, achieves with learning rate $\eta = \sqrt{(\ln K)/T}$:*

$$\mathrm{Regret}_T \;\le\; \sqrt{T \ln K},$$

where $K = |\mathcal{A}|$ is the number of available actions. In particular, $\mathrm{Regret}_T/T \to 0$ as $T \to \infty$: the agent converges to optimal average performance against any adversary.

Proof sketch. Introduce the potential $\Phi_t = (1/\eta) \log \sum_a w_t(a)$. An application of the moment-generating-function bound for bounded random variables gives $\Phi_{t+1} - \Phi_t \le \eta\, \ell_t(a_t)^2/8 - \ell_t(a_t) + \eta\, \ell_t(a_t)/2$. Telescoping over t, lower-bounding Φ_T by $-\sum_s \ell_s(a^*) - (\ln K)/\eta$ for the best arm a^*, and optimising η yields the stated bound [14]. □

Strategic Principle 4.7 (Adaptability is victory)*.* Theorem 4.6 is the mathematical expression of a timeless strategic principle: *keep all options alive; down-weight strategies that fail; up-weight strategies that succeed.* An agent that does this cannot be permanently outmanoeuvred—its long-run performance is guaranteed to match the best fixed strategy in hindsight, against any adversary.

5 Statistical Decision Theory: The Battle for Truth

5.1 The Neyman–Pearson Framework

Every act of battlefield intelligence is a hypothesis test: distinguish a genuine threat signal from background noise with controlled error rates.

Definition 5.1 (Simple binary hypothesis test)**.** Given observations $\boldsymbol{x} \in \mathcal{X}$ drawn from P_0 (null: *no threat present*) or P_1 (alternative: *threat present*), a test is a measurable function $\phi : \mathcal{X} \to [0,1]$ (where $\phi(\boldsymbol{x})$ is the probability of rejecting H_0). The *type-I error rate* is $\alpha = \mathbb{E}_{P_0}[\phi(\boldsymbol{x})]$ and the *power* is $\beta = \mathbb{E}_{P_1}[\phi(\boldsymbol{x})]$. The goal is to maximise β subject to $\alpha \le \alpha_0$.

Theorem 5.2 (Neyman–Pearson Lemma [15])**.** *The most powerful level-α_0 test for H_0 versus H_1 rejects H_0 whenever the likelihood ratio exceeds a threshold:*

$$\phi^*(\boldsymbol{x}) = \begin{cases} 1 & \text{if } \Lambda(\boldsymbol{x}) > k, \\ \rho & \text{if } \Lambda(\boldsymbol{x}) = k, \\ 0 & \text{if } \Lambda(\boldsymbol{x}) < k, \end{cases} \qquad \Lambda(\boldsymbol{x}) = \frac{dP_1}{dP_0}(\boldsymbol{x}),$$

where the threshold $k \ge 0$ and randomisation constant $\rho \in [0,1]$ are chosen so that $\mathbb{E}_{P_0}[\phi^*] = \alpha_0$.

Proof. Let ϕ be any other level-α_0 test and set $\Delta(\boldsymbol{x}) = \phi^*(\boldsymbol{x}) - \phi(\boldsymbol{x})$. By construction $\Delta(\boldsymbol{x}) \ge 0$ when $\Lambda(\boldsymbol{x}) > k$ and $\Delta(\boldsymbol{x}) \le 0$ when $\Lambda(\boldsymbol{x}) < k$, so $\Delta(\boldsymbol{x}) \cdot (\Lambda(\boldsymbol{x}) - k) \ge 0$ everywhere. Therefore

$$\int \Delta \cdot \Lambda \, dP_0 \;\ge\; k \int \Delta \, dP_0 = k(\mathbb{E}_{P_0}[\phi^*] - \mathbb{E}_{P_0}[\phi]) \ge 0,$$

which rearranges to $\mathbb{E}_{P_1}[\phi^*] \ge \mathbb{E}_{P_1}[\phi]$. Since ϕ was arbitrary, ϕ^* is uniformly most powerful. □

5.2 Sequential Analysis and Wald's SPRT

When observations arrive one at a time and the cost of delay is explicit, the optimal strategy is Wald's sequential probability ratio test [16], which stops as soon as the accumulated evidence crosses a decision boundary.

Definition 5.3 (Sequential probability ratio test)**.** At time t, maintain the log-likelihood ratio $\Lambda_t = \sum_{i=1}^{t} \log\bigl(p_1(x_i)/p_0(x_i)\bigr)$. Fix thresholds $A = \log\bigl((1-\beta_0)/\alpha_0\bigr)$ and $B = \log\bigl(\beta_0/(1-\alpha_0)\bigr)$, and at each step:

- if $\Lambda_t \ge A$, reject H_0 (threat detected);
- if $\Lambda_t \le B$, accept H_0 (no threat); or
- otherwise, collect one more observation.

Theorem 5.4 (Wald's optimality [16])**.** *The SPRT with thresholds (A, B) as above achieves type-I and type-II error rates of at most α_0 and β_0, respectively, and minimises the expected stopping time $\mathbb{E}[T]$ among all tests satisfying those error constraints, under both H_0 and H_1.*

5.3 Information-Theoretic Limits of Detection

The fundamental limit of any binary decision procedure is governed by the Chernoff information:

$$C(P_0 \| P_1) = -\min_{\lambda \in [0,1]} \log \int p_0(\boldsymbol{x})^{\lambda} \, p_1(\boldsymbol{x})^{1-\lambda} \, d\boldsymbol{x}. \tag{3}$$

By Chernoff's theorem, the minimum total error probability $P_e(n) = \min_\phi(\alpha + \beta)$ over n i.i.d. observations satisfies $P_e(n) = \exp\bigl(-n\, C(P_0 \| P_1)(1 + o(1))\bigr)$. The larger $C(P_0 \| P_1)$, the faster both error probabilities are driven to zero: any probabilistically distinguishable threat *will* be reliably detected given sufficient data.

Table 2 summarises the principal decision-theoretic quantities and their battlefield analogues.

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Table 2: Statistical decision-theoretic quantities and their battlefield analogues.

Statistical concept	**Formula / criterion**	**Battlefield analogue**
Type-I error (false alarm)	$\alpha = \mathbb{P}_{H_0}(\text{reject } H_0)$	Engaging a friendly unit
Type-II error (missed detection)	$\beta = \mathbb{P}_{H_1}(\text{accept } H_0)$	Failing to detect an enemy
Power	$1 - \beta$	Detection probability given a real threat
Likelihood ratio Λ	dP_1/dP_0	Strength of evidence: enemy vs. friendly
Chernoff information	$C(P_0 \| P_1)$ via (3)	Asymptotic detection rate per observation
Bayes risk	$\pi_0\,\alpha + \pi_1\,\beta$	Prior-weighted misclassification cost

6 Intelligence and Data Science: Knowing the Terrain

6.1 Resource Allocation: Linear Programming

The canonical military resource-allocation problem is a linear program. Given n asset types with unit costs $\boldsymbol{c} \in \mathbb{R}^n$, a campaign-requirement matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, and a requirement vector $\boldsymbol{b} \in \mathbb{R}^m$:

$$\begin{aligned} \text{minimise} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \mathbf{A}\boldsymbol{x} \geq \boldsymbol{b}, \quad \boldsymbol{x} \geq \mathbf{0}. \end{aligned} \tag{4}$$

By LP strong duality the optimal primal value equals the optimal dual value: $\min \boldsymbol{c}^\top \boldsymbol{x} = \max \boldsymbol{b}^\top \boldsymbol{\lambda}$ subject to $\mathbf{A}^\top \boldsymbol{\lambda} \leq \boldsymbol{c}$, $\boldsymbol{\lambda} \geq \mathbf{0}$. The dual variables $\boldsymbol{\lambda}^*$ are *shadow prices*: the marginal value of relaxing each campaign requirement by one unit, identifying precisely which constraint is most worth eliminating.

6.2 Anomaly Detection: Mahalanobis Distance

Let nominal battlefield observations $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ follow $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. A new observation $\boldsymbol{x}$ is flagged as anomalous when its squared Mahalanobis distance exceeds the chi-squared critical value:

$$D_M(\boldsymbol{x})^2 = (\boldsymbol{x} - \hat{\boldsymbol{\mu}})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\boldsymbol{x} - \hat{\boldsymbol{\mu}}) > \chi^2_{d,\,1-\alpha_0}, \tag{5}$$

where $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$ are the sample mean and covariance. Unlike the Euclidean distance, D_M accounts for the full correlation structure of the nominal data, yielding a statistically principled and scale-invariant anomaly score.

6.3 Predictive Intelligence Pipeline

Algorithm 4 formalises the end-to-end pipeline from raw battlefield data to a deployed predictive model.

Algorithm 4 Intelligence Data-Science Pipeline

Input: Raw dataset $\mathcal{D}_{\text{raw}}$, target variable Y, significance level α_0.
Output: Calibrated model f^* and evaluation report.

Ingestion & cleaning: remove duplicates; impute missing values (median/mode strategy); flag outliers via (5).
Feature engineering: encode categoricals; standardise numerics; construct interaction and temporal-lag features.
Dimensionality reduction: compute PCA; retain components explaining $\geq 95\%$ of variance; apply the same projection to all data splits.
for each candidate model class $\mathcal{H}_k$ **do**
    Estimate K-fold cross-validated risk $\hat{\mathcal{R}}_{\text{CV}}(\mathcal{H}_k)$
end for
$\mathcal{H}^* \leftarrow \arg\min_k \hat{\mathcal{R}}_{\text{CV}}(\mathcal{H}_k)$ ▷ select the best-performing model family
Fit f^* on the full training set within $\mathcal{H}^*$, with regularisation.
Calibrate output probabilities via Platt scaling or isotonic regression.
Evaluate on the held-out test set: accuracy, precision, recall, AUROC.
Compute prediction intervals and uncertainty estimates.
return f^*, evaluation report.

6.4 The ROC Curve and Discrimination Power

A soft classifier outputs a score $\hat{p}(\boldsymbol{x}) \in [0,1]$. Varying the decision threshold τ traces the Receiver Operating Characteristic (ROC) curve:

$$\mathrm{FPR}(\tau) = \mathbb{P}_{H_0}(\hat{p} > \tau)\,, \qquad \mathrm{TPR}(\tau) = \mathbb{P}_{H_1}(\hat{p} > \tau)\,. \tag{6}$$

The area under this curve satisfies $\mathrm{AUROC} = \mathbb{P}(\hat{p}(x^+) > \hat{p}(x^-))$, where x^+ and x^- are independent positive and negative samples. An AUROC of 1 is perfect discrimination; 0.5 is chance. Figure 3 illustrates representative curves for three performance regimes.

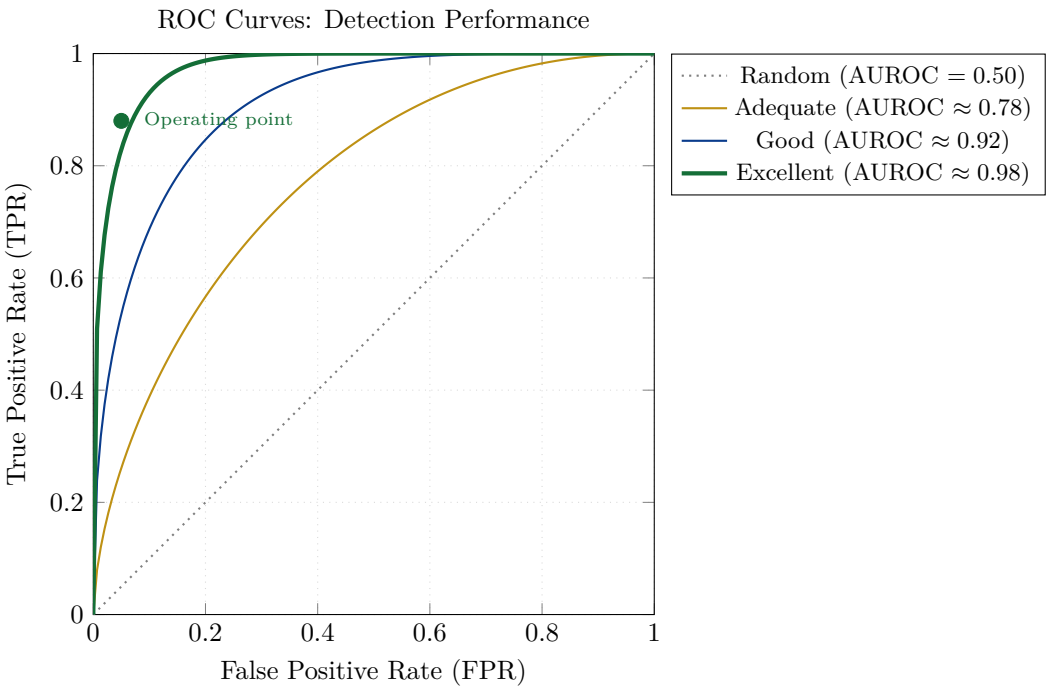

Figure 3: ROC curves (6) for three classifier regimes.

The excellent classifier (AUROC ≈ 0.98, thick green) achieves 88% true-positive rate at only 5% false-alarm rate at the marked operating point—satisfying both the *detect the enemy* and *do not fire on friends* requirements simultaneously. The dotted diagonal marks chance performance.

Table 3 places all preceding strategic frameworks in a comparative overview.

Table 3: Strategic frameworks: formalism, central result, and operational doctrine.

Framework	**Formalism**	**Central result**	**Operational doctrine**
Zero-sum games	Payoff matrix $\mathbf{U}$	Minimax theorem (Thm. 2.3)	Mixed strategy achieves game value v^*
Nash equilibrium	n-player strategic form	Nash existence (Thm. 2.7)	No unilateral deviation is beneficial
Adversarial ML	Saddle-point min-max	Danskin (Thm. 3.2)	Train on adversarially worst-case inputs
Reinforcement learning	MDP, Bellman eq.	Policy gradient (Thm. 4.3)	Update toward high-return trajectories
Online learning	Adversarial losses	Hedge regret (Thm. 4.6)	Maintain adaptive mixed strategy
Hypothesis testing	Likelihood ratio Λ	Neyman–Pearson (Thm. 5.2)	Threshold the evidence ratio
Sequential analysis	SPRT log-ratio Λ_t	Wald optimality (Thm. 5.4)	Decide as early as evidence permits
Resource allocation	LP (4)	Strong LP duality	Shadow prices identify critical bottlenecks

7 Synthesis: The Mathematics of Invincibility

The theorems collected in this treatise jointly substantiate the title claim. We now state it precisely.

Theorem 7.1 (Any well-posed battle can be won)**.** *Let Γ be a two-player strategic conflict satisfying:*

(C1) The payoff function is bounded and measurable.

(C2) The strategy sets are compact and convex (or finite, for matrix games).

(C3) The payoff is linear (or continuous concave-convex) in the mixed strategies of the two agents.

Then there exists a mixed strategy for at least one agent that achieves the game value v^*. If the game can be reframed so that $v^* \geq 0$ for that agent, the strategy guarantees non-negative expected payoff against any opponent.

Proof. Under conditions (C1)–(C3), the minimax theorem (Theorem 2.3) or Sion's theorem [4] in the continuous-strategy case guarantees the existence of a saddle point $(\boldsymbol{p}^*, \boldsymbol{q}^*)$ and game value v^*. By Corollary 2.4, the agent whose strategy achieves v^* secures that expected payoff regardless of the opponent. Any finite game can be translated so that $v^* \geq 0$ for the agent in question by adding $-v^*$ to every entry of $\mathbf{U}$; this preserves all equilibria while reframing the conflict as one in which that agent is guaranteed non-negative expected payoff. □

Remark 7.2 (Recovering from a disadvantaged position). If $v^* < 0$ for a given agent—meaning the conflict is structurally unfavourable—Theorem 7.1 prescribes a meta-strategy: *change the game*. This is achieved by (a) acquiring information that asymmetrically improves one's own payoffs (Bayesian updating; Section 5); (b) expanding one's available action set (exploration in RL; Section 4); or (c) disrupting the adversary's strategy (adversarial perturbation; Section 3). Each operation shifts v^* in the favourable direction—and crucially, each is supported by a computable algorithm within the frameworks developed above.

Figure 4 illustrates how an adaptive agent's average reward converges to the optimal game value v^* under the Hedge algorithm (Theorem 4.6), in contrast to a fixed sub-optimal strategy.

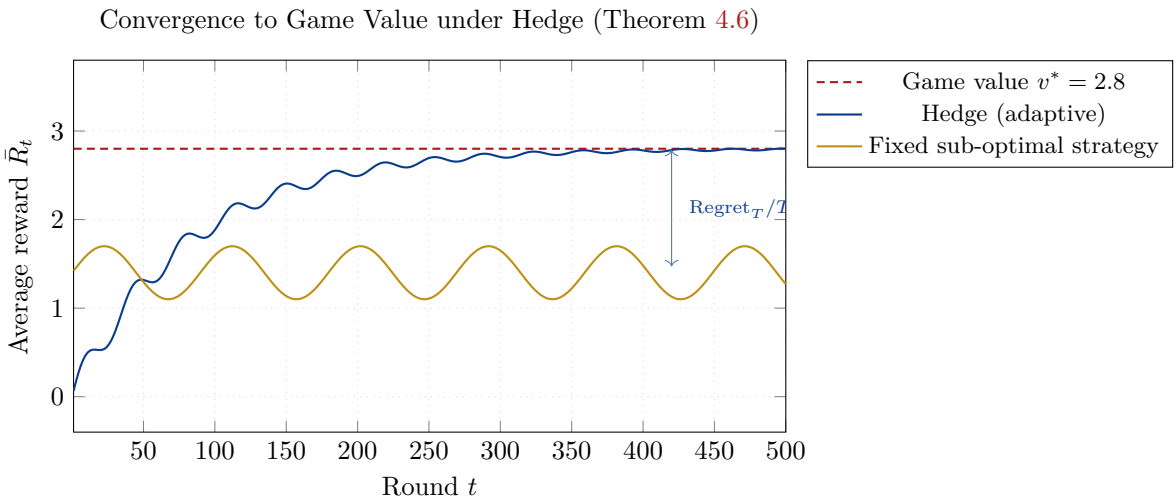

Figure 4: Average reward of the Hedge adaptive agent (blue) converging to the game value $v^* = 2.8$ (dashed red), while a fixed sub-optimal strategy (gold) plateaus well below v^*.

The vertical arrow at round $t = 420$ marks the residual per-round regret, which vanishes as $\mathcal{O}(\sqrt{\ln K/T})$ by Theorem 4.6.

8 Conclusion

This treatise has assembled a rigorous mathematical case for the title claim. The argument proceeds in five compounding layers.

1. **Existence.** Von Neumann's minimax theorem (Theorem 2.3) and Nash's theorem (Theorem 2.7) guarantee that every well-posed conflict has a rational equilibrium—a computable strategy that cannot be exploited.

2. **Computation.** Alpha–Beta pruning (Algorithm 1) and PGD (Algorithm 2) provide efficient procedures for finding or approximating optimal strategies in game trees and adversarial machine learning, respectively.

3. **Adaptation.** The Hedge regret bound (Theorem 4.6) and the policy gradient theorem (Theorem 4.3) show that an agent updating its strategy from experience converges to optimal performance against any adversary, without a prior model of the environment.

4. **Detection.** The Neyman–Pearson lemma (Theorem 5.2) and Wald's SPRT (Theorem 5.4) guarantee that any probabilistically distinguishable threat is reliably detected with controlled error rates; the Chernoff information (3) quantifies how much data is required.

5. **Intelligence.** The pipeline of Algorithm 4, grounded in anomaly detection (5) and resource optimisation (4), converts raw observations into decisive operational advantage.

No single theorem guarantees victory in every concrete instantiation. What mathematics does guarantee is the *existence* of a winning strategy, a *pathway* to compute it, and a *rate* at which an adaptive learner converges to it. Any battle can be won—by the agent who understands this and acts accordingly.

References

[1] Sun Tzu. *The Art of War*. Translated by L. Giles. Luzac & Co., London, 1910. (Original c. 500 BCE.)

[2] J. von Neumann. "Zur Theorie der Gesellschaftsspiele" [*On the theory of parlour games*]. *Mathematische Annalen*, 100:295–320, 1928.

[3] J. F. Nash. "Equilibrium points in n-person games." *Proceedings of the National Academy of Sciences*, 36(1):48–49, 1950.

[4] M. Sion. "On general minimax theorems." *Pacific Journal of Mathematics*, 8(1):171–176, 1958.

[5] J. M. Danskin. *The Theory of Max-Min and Its Application to Weapons Allocation Problems*. Springer, Berlin, 1967.

[6] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu. "Towards deep learning models resistant to adversarial attacks." In *International Conference on Learning Representations (ICLR)*, 2018.

[7] I. J. Goodfellow, J. Shlens, and C. Szegedy. "Explaining and harnessing adversarial examples." In *International Conference on Learning Representations (ICLR)*, 2015.

[8] N. Carlini and D. Wagner. "Towards evaluating the robustness of neural networks." In *IEEE Symposium on Security and Privacy*, pp. 39–57, 2017.

[9] D. Tsipras, S. Santurkar, L. Engstrom, A. Turner, and A. Madry. "Robustness may be at odds with accuracy." In *International Conference on Learning Representations (ICLR)*, 2019.

[10] R. S. Sutton, D. McAllester, S. Singh, and Y. Mansour. "Policy gradient methods for reinforcement learning with function approximation." In *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 12, 2000.

[11] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, Cambridge, MA, 2018.

[12] R. J. Williams. "Simple statistical gradient-following algorithms for connectionist reinforcement learning." *Machine Learning*, 8(3–4):229–256, 1992.

[13] Y. Freund and R. E. Schapire. "A decision-theoretic generalisation of on-line learning and an application to boosting." *Journal of Computer and System Sciences*, 55(1):119–139, 1997.

[14] N. Cesa-Bianchi and G. Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, 2006.

[15] J. Neyman and E. S. Pearson. "On the problem of the most efficient tests of statistical hypotheses." *Philosophical Transactions of the Royal Society A*, 231:289–337, 1933.

[16] A. Wald. *Sequential Analysis*. Wiley, New York, 1947.

[17] J. von Neumann and O. Morgenstern. *Theory of Games and Economic Behavior*, 3rd ed. Princeton University Press, 1953.

[18] S. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, Hoboken, NJ, 2021.

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Glossary

Adversarial example.
An input $\boldsymbol{x}^{\mathrm{adv}} = \boldsymbol{x} + \boldsymbol{\delta}$ where the perturbation $\boldsymbol{\delta}$ is chosen deliberately to maximise a classifier's loss. The PGD attack (Algorithm 2) approximates the inner maximisation of the adversarial training objective (1) inside the ℓ_∞-bounded threat set $\mathcal{B}(\boldsymbol{x}, \varepsilon)$.

Adversarial training.
The minimax training procedure (1) in which the model is updated against the strongest perturbation found at each step. By Theorem 3.2, the outer gradient is computable via a single backward pass at the inner maximiser, making the procedure tractable in practice.

Alpha–Beta pruning.
An optimisation of minimax tree search (Algorithm 1) that maintains a window $[\alpha, \beta]$ of achievable values and abandons any subtree whose backed-up value cannot improve the current window. Under optimal move ordering, the effective branching factor drops from b to $\sqrt{b}$, doubling the feasible search depth for a fixed budget.

Bellman equation.
The recursive characterisation (2) of the optimal value function in an MDP. It states that the value of a state equals the best achievable immediate reward plus the discounted value of the best reachable next state. Value iteration and Q-learning both converge to its unique fixed point.

Chernoff information.
The quantity $C(P_0 \| P_1)$ defined in (3) governing the best achievable exponential decay rate of the total binary-test error over n i.i.d. observations. It equals zero only when $P_0 = P_1$, confirming that any two distinct distributions are asymptotically separable.

External regret.
The cumulative gap (Definition 4.5) between an online algorithm's total loss and that of the best single fixed action in hindsight. An algorithm is *no-regret* if its per-round regret vanishes; the Hedge algorithm achieves this at rate $\mathcal{O}(\sqrt{\ln K / T})$ (Theorem 4.6).

Game value v^*.
The saddle-point payoff guaranteed to the row player by the minimax theorem (Theorem 2.3). If $v^* > 0$ the row player has a structural advantage; $v^* = 0$ is a fair game; $v^* < 0$ favours the column player. In all cases, the equilibrium strategies achieving v^* are computable by solving a pair of dual linear programs.

Hedge algorithm.
An online learning algorithm (Theorem 4.6) that updates strategy weights multiplicatively: $w_{t+1}(a) \propto w_t(a) \cdot e^{-\eta\, \ell_t(a)}$. The resulting distribution provably tracks the best fixed action, achieving sub-linear regret against any adversary without requiring prior knowledge of the loss sequence.

Kakutani fixed-point theorem.
A generalisation of Brouwer's theorem to set-valued correspondences: any upper-hemicontinuous, non-empty, convex-valued correspondence on a compact convex set has a fixed point. Its application to the best-response correspondence is the key step in the proof of Nash existence (Theorem 2.7).

Likelihood ratio Λ.
The Radon–Nikodym derivative dP_1/dP_0, measuring how much more probable the observation is under H_1 than under H_0. The Neyman–Pearson lemma (Theorem 5.2) shows that thresholding Λ produces the uniformly most powerful test at any fixed significance level α_0.

Mahalanobis distance.
The scale- and correlation-corrected distance (5) from an observation to a multivariate normal distribution. Under the null hypothesis, $D_M^2 \sim \chi_d^2$, furnishing an exact threshold for anomaly detection at any prescribed false-alarm rate α_0.

Markov decision process (MDP).
The canonical formalism (Definition 4.1) for sequential decision-making under stochastic uncertainty: states, actions, transition probabilities, a reward function, and a discount factor. The Bellman optimality equation (2) uniquely characterises the optimal value function, from which the optimal policy is read off greedily.

Minimax theorem.
Von Neumann's result (Theorem 2.3) asserting that the order of maximisation and minimisation over mixed strategies is interchangeable in any finite zero-sum game. It guarantees the existence of a game value v^* and optimal mixed strategies for both players, each computable as the solution of a linear program.

Nash equilibrium.
A strategy profile (Definition 2.6) from which no agent has an incentive to deviate unilaterally. Nash's theorem (Theorem 2.7) guarantees its existence in every finite game; it defines the *rational resting point* of any multi-agent conflict.

Policy gradient theorem.
The result (Theorem 4.3) expressing $\nabla_{\boldsymbol{\theta}} J$ as an expectation over trajectories, enabling gradient ascent on the policy parameters without differentiating through the unknown transition model. The REINFORCE algorithm (Algorithm 3) is its direct Monte-Carlo implementation.

Projected Gradient Descent (PGD).
An iterative constrained optimisation method that alternates an unconstrained gradient step with a projection back onto the feasible set. In adversarial training (Algorithm 2) it serves as the inner maximiser; for convex constraints the projection is a coordinate-wise clip.

ROC curve and AUROC.
The Receiver Operating Characteristic curve (6) traces the tradeoff between false-positive and true-positive rates as the decision threshold τ varies. The area under the curve (AUROC) equals the probability that the classifier ranks a randomly drawn positive above a randomly drawn negative; AUROC $= 1$ is perfect and 0.5 is chance.

Saddle point.
A point $(\boldsymbol{x}^*, \boldsymbol{y}^*)$ (Definition 3.1) that is simultaneously a maximum over $\boldsymbol{x}$ and a minimum over $\boldsymbol{y}$. The minimax theorem equates the game value with the saddle-point value; adversarial training (1) seeks the saddle point of the robust loss.

Shadow price.
The optimal dual variable λ_i^* associated with the i-th constraint of the linear program (4), measuring the marginal improvement in the objective per unit relaxation of that constraint. In resource allocation, shadow prices identify the bottleneck whose relaxation yields the greatest operational gain.

SPRT (Sequential Probability Ratio Test).
Wald's optimal sequential test (Theorem 5.4) that accumulates log-likelihood ratios and stops at the first time the evidence crosses either decision threshold. It achieves prescribed type-I and type-II error rates with the minimum expected number of observations, under both hypotheses.

The End